



Comparative virulence of *Beauveria bassiana* isolates against lepidopteran pests of vegetable crops

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ABSTRACT

Forty-three isolates of the entomopathogenic fungus *Beauveria bassiana* were screened for virulence against second-instar larvae of diamondback moth (*Plutella xylostella*) (DBM), European corn borer (*Ostrinia nubilalis*) (ECB), corn earworm (*Helioverpa zea*) (CEW), and fall armyworm (*Spodoptera frugiperda*) (FAW); 30 of these isolates were tested against beet armyworm (*Spodoptera exigua*) (BAW). Highly virulent isolates were also tested against black cutworm (*Agrotis ipsilon*) (BCW), and the most virulent isolate was also assayed against imported cabbage worm (*Pieris rapae*) (ICW) and cabbage looper (*Trichoplusia ni*) (CL). All lepidopteran species tested were susceptible to *B. bassiana*. Corn earworm and beet armyworm were most susceptible to fungal infection, and fall armyworm was least susceptible. Limited testing suggested low susceptibility of black cutworm and cabbage looper. *B. bassiana* isolate 1200 exhibited virulence against all pest species greater than or equal to commercial strain GHA of *B. bassiana* currently registered in the USA as BotaniGard®. In assays in which larvae were topically sprayed and maintained on the treated substrate for 24 h at 100% relative humidity, 6-day (25 °C) median lethal concentrations (LC₅₀s) of this isolate against CEW, BAW, DBM, FAW, ICW, ECB, CL, and BCW were 4, 5, 7, 11, 12, 98, 125, and 273 conidia/mm², respectively. The respective LC₅₀s of commercial strain GHA against these pest species were 9, 67, 97, 1213, 29, 1668, 541, and 3504 conidia/mm². Use of LC₅₀ versus median lethal concentration ratios (comparing LC₅₀s of each isolate to a “standard” strain) generated similar rankings of isolate virulence. Results from parametric ANOVAs of log LC₅₀ values followed by Tukey HSD multiple comparisons tests and those from Kruskal–Wallis nonparametric analyses followed by sequential Bonferroni tests for means comparisons were nearly identical.

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1. Introduction

Recent advances in fungal production, stabilization, formulation, and application have led the way toward commercialization of a large number of new fungus-based biopesticide products (Wraight et al., 2001; Faria and Wraight, 2007). Despite this growth in the mycopesticide industry, however, the market for these products has been slow to develop and remains an exceedingly small percentage of insecticide sales worldwide. The reasons for this are numerous and diverse, but one of the most difficult challenges these agents face in competing with chemical insecticides relates to their generally narrow host ranges. Typically, crop damage results not from the activities of a single pest but from a

complex of pest species, and individual strains of fungal pathogens are rarely virulent against more than one insect species or a few closely related species infesting a crop (Glare and Milner, 1991; Hajek, 1997). It is generally not economically feasible for growers to target each member of a pest complex with a different product or for commercial developers to register, produce, and market myriad strains of fungal pathogens.

This problem may be overcome by: (1) integration of fungal pathogens with other control agents and practices, especially in bio-intensive IPM systems designed to conserve natural enemy complexes, (2) formulation of fungi to enhance efficacy, and (3) identification of fungal strains with broader host spectra. The first two approaches represent important and growing areas of research in the field of applied insect mycopathology. The third approach has received somewhat less attention. Extensive culture collections have been amassed, but screening of the innumerable isolates against even a single pest species represents a daunting and costly task with no assurance of success; moreover, useful results from largely unsuccessful efforts are rarely published.

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The primary objective of this study was to identify one or more strains of *Beauveria bassiana* virulent against a broad range of lepidopteran pests of vegetable crops. Forty-three isolates were tested against diamondback moth (DBM) (*Plutella xylostella*), European corn borer (ECB) (*Ostrinia nubilalis*), corn earworm (CEW) (*Helicoverpa zea*), and fall armyworm (FAW) (*Spodoptera frugiperda*), and thirty of these isolates were also tested against beet armyworm (BAW) (*Spodoptera exigua*). The most virulent isolate and other promising isolates identified in the initial screening assays were also tested against black cutworm (*Agrotis ipsilon*), imported cabbageworm (*Pieris rapae*) and cabbage looper (*Trichoplusia ni*). This collaborative effort between the former Mycotech Corporation of Butte, Montana (now Laverlam International) and USDA-ARS had an additional objective of identifying an isolate of *B. bassiana* with a broader lepidopteran host range than the commercial strain GHA. At the time of this study, Mycotech was investigating the potential of *B. bassiana* for control of a number of major insect pests/pest complexes, including armyworms, cornborers, fleas, and boll weevils, and therefore the sample of isolates selected for initial screening, rather than being comprised solely of known lepidopteran pathogens, was heavily weighted toward isolates associated with these diverse pests. A secondary objective was to investigate various aspects of the experimental design and statistical analysis of fungal screening bioassays. LC_{50} s of insect pathogens generally exhibit high assay-to-assay variability (Robertson et al., 1995), and protocols designed to reduce this variability have included calculation of potency or lethal dose ratios based on use of “standard” strains, but the effectiveness of this approach has been questioned (Wraight et al., 1998). Distributions of LC_{50} values are also potentially non-normal, which can complicate statistical analyses. Conduct of a large number of assays that included a designated standard enabled investigation of these issues.

2. Materials and methods

2.1. Test insects

Insects were obtained as eggs from cooperators at other USDA-ARS laboratories, Boyce Thompson Institute (Ithaca, NY), or from a commercial supplier (Table 1). All larvae were reared to second instar on wheat germ/soybean flour/agar-based diets (Southland Products, Inc., Lake Village, AR) dispensed in 16 oz. (473 ml) Styrofoam cups or standard Petri dishes (90 mm, disposable polystyrene). All diets included aureomycin to inhibit bacterial growth, but no anti-fungal agents were added. After preparation, diets were stored at 4 °C. Eggs or egg masses were distributed on the diet sur-

face in the plastic cups (approximately 200 eggs/cup) and incubated at 15–30 °C, depending on desired timing of hatch. The incubator was set to provide a 16 h daily photoperiod. Larvae were reared in the cups until molt to the second instar.

Early second-instar larvae were selected from the cups and placed in replicate batches of 12–15 on lids of standard Petri dishes lined with filter paper moistened with distilled water (dH₂O). The lid was then covered with the Petri dish bottom. After selection of the desired number of insect batches, dishes were randomly assigned to treatments, labeled, and held at 10 °C until treated (no longer than 4 h).

2.2. Fungal isolates and preparations

All *B. bassiana* isolates were selected from the USDA-ARS Collection of Entomopathogenic Fungal Cultures (ARSEF) or from the Mycotech holdings (Table 2). All assays used dry conidia from solid substrate fermentations conducted by Mycotech, using methods and ingredients based on Bradley et al. (1992). Spore preparations were received from Mycotech in the form of unformulated technical powders with >85% viability. These were stored in sealed containers at 4 °C for the duration of the study. All but a few conidial powders retained initial viability for the duration of the study; the least stable powders retained at least 70% viability (Tables 3–7).

2.3. Screening bioassay protocol

The basic measure of virulence generated in this study was the median lethal concentration (LC_{50}) expressed as viable conidia/mm² of sprayed target surface and based on mortality recorded 6 days post-inoculation. A screening assay of a selected fungal isolate comprised three doses of conidia, each applied to two or three of the replicate batches of larvae described above (24–36 larvae/dose, 72–108 larvae/assay).

Conidial suspensions were prepared by mixing 10–60 mg of conidial powder (depending on the targeted concentration) in 20 ml 0.01% aqueous (dH₂O) Silwet L-77 (Loveland Industries, Greeley, CO) in a 50-ml plastic centrifuge tube, adding 1 g glass beads (2 mm diam.), and shaking for 15 min. on a mechanical wrist action shaker (Burrell Scientific, Pittsburgh, PA, Model 75) set at 380 oscillations per minute. In most cases, the resulting suspension was serially diluted 5- and 25-fold by pipetting 5 ml aliquots into 20 ml aqueous Silwet in centrifuge tubes. In assays of highly virulent isolates, however, it was necessary to dilute the entire 20 ml suspension 10- or 25-fold prior to preparing the final dilution series. The three dilutions, representing high, medium, and low doses

Table 1
Lepidoptera species included in bioassays.

Species	Family (subfamily)	Laboratory colony origin	Supplier
Diamondback moth <i>Plutella xylostella</i> (Linnaeus)	Plutellidae (Plutellinae)	Geneva, New York, 1988	NYS Agricultural Experiment Station, Geneva, NY
European corn borer <i>Ostrinia nubilalis</i> (Hübner)	Pyralidae (Pyraustinae)	Central Iowa	USDA-ARS Corn Insects and Crop Genetics Research Unit, Ames, IA
Corn earworm <i>Helicoverpa zea</i> (Boddie)	Noctuidae (Heliothinae)	Stoneville, Mississippi, infused biennially with wild Minnesota stock	French Agricultural Research Co., Lamberton, MN
Fall armyworm <i>Spodoptera frugiperda</i> (J.E. Smith)	Noctuidae (Amphipyrrinae)	Minnesota, 1998	French Agricultural Research Co., Lamberton, MN
Beet armyworm <i>Spodoptera exigua</i> (Hübner)	Noctuidae (Amphipyrrinae)	Stoneville, Mississippi	USDA-ARS Biological Control and Mass Rearing Research Unit, Stoneville, MS
Black cutworm <i>Agrotis ipsilon</i> (Hufnagel)	Noctuidae (Noctuinae)	Manhattan, Kansas	French Agricultural Research Co., Lamberton, MN
Cabbage looper <i>Trichoplusia ni</i> (Hübner)	Noctuidae (Plusiinae)	Riverhead, Long Island, New York, 1977	Boyce Thompson Institute, Ithaca, NY
Imported Cabbageworm <i>Pieris rapae</i> (Linnaeus)	Pieridae (Pierinae)	Ithaca, NY, infused annually with wild NY stock	Boyce Thompson Institute, Ithaca, NY

Table 2Identification of *Beauveria bassiana* isolates assayed against Lepidoptera larvae.

Isolate (Holding) ^a	Insect host of origin	Reisolation hosts	Site and date of origin
959 (ARSEF)	<i>Spodoptera frugiperda</i>	–	Londrina, Paraná, Brazil, 1983
1153 (ARSEF)	<i>Sitona discoideus</i>	–	Jramma, Morocco, 1984
1512 (ARSEF)	<i>Spodoptera littoralis</i>	–	La Minière, France, 1984
1513 (ARSEF)	<i>Spodoptera littoralis</i>	–	La Minière, France, 1984
1835 (ARSEF)	<i>Anthonomus grandis</i>	–	Brazil, 1985
1966 (ARSEF)	<i>Anthonomus grandis</i>	–	Campinas, São Paulo, Brazil, 1983
2559 (ARSEF)	<i>Anthonomus grandis</i>	–	Piracicaba, São Paulo, Brazil, 1988
2638 (ARSEF)	<i>Anthonomus grandis</i>	–	Piracicaba, São Paulo, Brazil, 1988
2945 (ARSEF)	<i>Anthonomus grandis</i>	–	Piracicaba, São Paulo, Brazil, 1988
3286 (ARSEF)	<i>Spodoptera littoralis</i>	–	Montpellier, Hérault, France, 1987
3288 (ARSEF)	<i>Spodoptera frugiperda</i>	–	Colima, Mexico, 1988
1003 (Mycotech)	Corn plant	–	Lamberton, Minnesota, USA, 1997
1005 (Mycotech)	Corn plant	–	Lamberton, Minnesota, USA, 1997
1008 (Mycotech)	Corn plant	–	Champaign, Illinois, USA, 1997
1009 (Mycotech)	Corn plant	–	Champaign, Illinois, USA, 1997
1012 (Mycotech)	Corn plant	–	Ames, Iowa, USA, 1997
1013 (Mycotech)	Corn plant	–	Ames, Iowa, USA, 1997
1014 (Mycotech)	Corn plant	–	Lafayette, Indiana, USA, 1997
1015 (Mycotech)	Corn plant	–	Lafayette, Indiana, USA, 1997
1019 (Mycotech)	Corn plant	–	Concord, Nebraska, USA, 1997
1020 (Mycotech)	Corn plant	–	Concord, Nebraska, USA, 1997
1022 (Mycotech)	Corn plant	–	Fargo, North Dakota, 1997
1023 (Mycotech)	Corn plant	–	Fargo, North Dakota, 1997
1055 (Mycotech)	<i>Diabrotica undecimpunctata</i> (GHA)	<i>Melanoplus sanguinipes</i> → <i>Bemisia tabaci</i> biotype B → <i>Ostrinia nubilalis</i>	Corvallis, Oregon, USA, 1977
1059 (Mycotech)	<i>Nezara viridula</i> (ARSEF 1054)	<i>Ctenocephalides felis</i>	Londrina, Paraná, Brazil, 1983
1060 (Mycotech)	<i>Heliothis virescens</i> (ARSEF 1314)	<i>Ctenocephalides felis</i>	La Minière, France, 1984
1067 (Mycotech)	<i>Leptinotarsa decemlineata</i> (ARSEF 252)	<i>Plutella xylostella</i> → <i>Myzus persicae</i> → <i>Ctenocephalides</i> sp.	Orono, Maine, USA, 1978
1075 (Mycotech)	<i>Blissus leucopterus</i> (ARSEF 792)	<i>Blissus leucopterus</i> → <i>Ctenocephalides felis</i>	Riley Co., Kansas, USA, 1981
1076 (Mycotech)	<i>Plutella xylostella</i> (ARSEF 2727)	<i>Ctenocephalides felis</i>	La Trinidad, Benguet, Philippines, 1989
1077 (Mycotech)	<i>Forficula auricularia</i> (ARSEF 3018)	<i>Ctenocephalides felis</i>	Brasília, Brazil, 1988
1082 (Mycotech)	<i>Choristoneura</i> sp. (ARSEF 1850)	<i>Plutella xylostella</i>	Terrace, British Columbia, Canada, 1983
1083 (Mycotech)	<i>Lymantria dispar</i> (ARSEF 3404)	<i>Plutella xylostella</i>	C&O Canal, Maryland, USA, 1991
1084 (Mycotech)	<i>Lymantria dispar</i> (ARSEF 3530)	<i>Plutella xylostella</i>	Piclic Run, Maryland, 1991
1085 (Mycotech)	<i>Plutella xylostella</i> (ARSEF 4525)	<i>Plutella xylostella</i>	Ontario Co., New York, USA, 1993
1086 (Mycotech)	<i>Plutella xylostella</i> (ARSEF 4543–single-spore from ARSEF 4534)	<i>Plutella xylostella</i>	Ontario Co., New York, USA, 1993
1200 (Mycotech)	<i>Plutella xylostella</i>		Proprietary
GHA (Mycotech)	<i>Diabrotica undecimpunctata</i> (ARSEF 201) ^b	<i>Melanoplus sanguinipes</i> → <i>Bemisia tabaci</i> biotype B	Corvallis, Oregon, USA, 1977
GMB6B (Mycotech)	<i>Lymantria dispar</i>	–	New York, USA, 1984
SA8 (Mycotech)	Soil in flea habitat	<i>Ctenocephalides felis</i>	1997
SC15 (Mycotech)	Soil in flea habitat	<i>Ctenocephalides felis</i>	1997
SC19 (Mycotech)	Soil in flea habitat	<i>Ctenocephalides felis</i>	1997
SE6 (Mycotech)	Soil in flea habitat	<i>Ctenocephalides felis</i>	1997
SF20 (Mycotech)	Soil in flea habitat	<i>Ctenocephalides felis</i>	1997

^a ARSEF: US Department of Agriculture, Agricultural Research Service, Collection of Entomopathogenic Fungus Cultures, Ithaca, NY; Mycotech Corp., Butte, MT (now Laverlam International Corp.).

^b Strain GHA originated from ARSEF 201 (Bradley et al., 1999), but it is genetically distinct from the currently held ARSEF 201 (S. Rehner, personal communication).

(application rates), were refrigerated (ca. 4 °C) prior to use; all suspensions were used within a few hours of preparation.

Spray applications were applied using a spray chamber fabricated after Burgerjon (1956). The chamber was fitted with an atomizing nozzle (Fluid Cap 2850 + Air Cap 70) mounted in a 1/4 J nozzle body (Spraying Systems Co., Wheaton, IL). The nozzle was connected to a regulator valve providing a constant airflow of 10 l/min. The target surface in the spray chamber was a rotating turntable (33 rpm), and spray deposition at the level of the target surface was approximately 0.01 µl/mm² (resulting from spraying of a 5 ml aliquot).

Dishes containing larvae were removed from refrigeration, covers were removed, larvae were brushed to the centers of the dishes, and the dishes were placed on the turntable (all at the same radial distance from the turntable center). An identical, clean Petri dish lid was also placed on the turntable as described above to sample spray deposition. The spray suspension was then fed into the spray

nozzle assembly. The process was repeated for each dose. During application of each isolate, a Petri dish containing freshly prepared 0.5% yeast extract agar (YEA) was included on the turntable to collect a sample for conidial viability determination. Treated dishes were removed from the chamber, provisioned with three small pieces of diet (5 mm cubes), re-covered, and sealed in small plastic bags to maintain high-humidity conditions. Viability plates were also sealed in plastic bags. The deposition sample plates were left uncovered until dry and were then covered and stored at room temperature.

All treated larvae and viability plates were incubated at 25 °C for 24 h (in the same incubator described above). After 24 h, 12 larvae from each dish were transferred to individual holding cells in a 12-well plate (22 mm well diam.) (BD Biosciences, Franklin Lakes, NJ) that had been previously provisioned with 5-mm cubes of diet; excess larvae were discarded. The 12-well plates were covered with the plastic lids provided by the manufacturer but modified

Table 3Virulence of 43 isolates of *Beauveria bassiana* against diamondback moth.

Isolate (% viability) ^a	Assays (n)	LC ₅₀ (log LC ₅₀ ± SE) rank ^{b,c}	Slope ± SE	Median LC ratio ± SE (rank) ^c	% Mortality (dose) C ± SE ^d
1200 (93)	3	7 (0.84 ± 0.33) 1	1.97 ± 0.12	12.99 ± 2.5 (2)	97 (15) 11 ± 8.0
1966 (85)	2	13 (1.11 ± 0.21) 2	1.04 ± 0.15	4.22 ± 1.8 (4)	86 (158) 5 ± 5.2
1084 (94)	2	24 (1.38 ± 0.47) 3	1.23 ± 0.51	14.47 ± 13.4 (1)	98 (2527) 13 ± 3.4
3288 (92)	2	25 (1.40 ± 0.45) 4	1.37 ± 0.30	3.18 ± 1.8 (5)	91 (145) 21 ± 6.8
1082 (91)	2	29 (1.46 ± 0.06) 5	1.36 ± 0.39	5.57 ± 3.4 (3)	90 (417) 13 ± 3.4
1075 (92)	2	55 (1.74 ± 0.07) 6	1.10 ± 0.04	0.97 ± 0.1 (10)	98 (870) 7 ± 3.1
SA8 (95)	2	59 (1.77 ± 0.13) 7	1.76 ± 0.70	0.93 ± 0.3 (11)	91 (1004) 7 ± 3.1
1153 (89)	2	59 (1.77 ± 0.19) 7	1.33 ± 0.18	1.09 ± 0.2 (8)	97 (593) 7 ± 7.3
3286 (91)	2	60 (1.78 ± 0.29) 9	1.40 ± 0.30	1.16 ± 0.3 (7)	65 (128) 21 ± 6.8
1083 (95)	3	61 (1.79 ± 0.17) 10	1.45 ± 0.19	2.98 ± 0.5 (6)	95 (986) 11 ± 2.9
1835 (80)	2	75 (1.88 ± 0.11) 11	1.58 ± 0.63	0.81 ± 0.3 (12)	53 (55) 3 ± 2.8
SF20 (96)	2	82 (1.91 ± 0.14) 12	1.52 ± 0.22	1.04 ± 0.2 (9)	98 (1315) 7 ± 2.7
GHA (92)	28	97 (1.99 ± 0.05) 13	1.36 ± 0.09	–	98 (577) 10 ± 1.5
1059 (95)	2	100 (2.00 ± 0.13) 14	1.23 ± 0.35	0.72 ± 0.01 (13)	90 (1441) 17 ± 7.5
2638 (92)	3	104 (2.02 ± 0.06) 15	1.65 ± 0.33	0.70 ± 0.2 (14)	86 (605) 16 ± 6.6
2945 (87)	2	111 (2.05 ± 0.06) 16	1.37 ± 0.31	0.68 ± 0.3 (15)	67 (205) 21 ± 6.8
2559 (89)	2	132 (2.12 ± 0.29) 17	1.05 ± 0.15	0.47 ± 0.3 (18)	94 (781) 5 ± 5.2
1055 (89)	2	148 (2.17 ± 0.17) 18	1.18 ± 0.32	0.55 ± 0.2 (16)	93 (1045) 11 ± 1.6
SE6 (91)	2	162 (2.21 ± 0.10) 19	1.55 ± 0.19	0.52 ± 0.1 (17)	98 (2463) 7 ± 2.8
1077 (93)	2	236 (2.37 ± 0.19) 20	1.69 ± 0.01	0.47 ± 0.2 (18)	97 (3069) 10 ± 7.1
1076 (93)	2	435 (2.64 ± 0.30) 21	0.97 ± 0.23	0.16 ± 0.1 (23)	89 (3648) 13 ± 7.6
1086 (93)	2	492 (2.69 ± 0.23) 22	1.56 ± 0.82	0.20 ± 0.1 (21)	87 (2437) 11 ± 5.3
SC19 (97)	2	630 (2.80 ± 0.17) 23	1.35 ± 0.46	0.40 ± 0.3 (20)	82 (2418) 15 ± 10.5
SC15 (97)	2	858 (2.93 ± 0.07) 24	1.43 ± 0.37	0.18 ± 0.1 (22)	85 (3888) 14 ± 10.6
1012 (95)	2	1382 (3.14 ± 0.25) 25	0.92 ± 0.20	0.09 ± 0.7 (25)	75 (5398) 15 ± 5.8
1512 (94)	1	1483 (3.17 ± –) 26	2.54 ± –	0.02 ± – (36)	81 (2935) 7 ± –
1009 (79)	2	1906 (3.28 ± 0.54) 27	1.44 ± 0.25	0.12 ± 0.1 (24)	80 (1986) 19 ± 2.1
1023 (94)	2	1926 (3.28 ± 0.04) 28	1.20 ± 0.30	0.05 ± 0.01 (28)	68 (4422) 6 ± 3.9
1067 (94)	2	2058 (3.31 ± 0.12) 29	1.85 ± 0.17	0.05 ± 0.03 (28)	93 (4715) 13 ± 3.4
1513 (93)	2	2343 (3.37 ± 0.04) 30	1.32 ± 0.14	0.02 ± 0.00 (36)	56 (2918) 5 ± 5.2
1085 (94)	2	3831 (3.58 ± 0.01) 31	0.66 ± 0.08	0.04 ± 0.03 (30)	53 (4266) 11 ± 5.3
1014 (93)	2	4068 (3.61 ± 0.35) 32	1.47 ± 0.90	0.04 ± 0.03 (30)	82 (4387) 15 ± 5.8
1003 (94)	2	4441 (3.65 ± 0.13) 33	1.36 ± 0.17	0.08 ± 0.04 (26)	49 (6086) 14 ± 7.2
1019 (93)	2	4747 (3.68 ± 0.31) 34	0.91 ± 0.01	0.02 ± 0.01 (36)	78 (5262) 10 ± 0.6
1020 (92)	2	4850 (3.69 ± 0.31) 35	0.72 ± 0.18	0.03 ± 0.01 (32)	69 (6120) 6 ± 3.9
1008 (81)	2	4922 (3.69 ± 0.26) 36	2.39 ± 1.53	0.03 ± 0.02 (32)	90 (5562) 17 ± 4.1
1060 (90)	1	5556 (3.74 ± –) 37	2.26 ± –	0.03 ± – (32)	61 (7020) 10 ± –
1022 (95)	2	5781 (3.76 ± 0.03) 38	2.10 ± 0.80	0.02 ± 0.01 (36)	65 (7053) 6 ± 3.6
1015 (89)	2	6260 (3.80 ± 0.16) 39	1.08 ± 0.11	0.03 ± 0.02 (32)	65 (5685) 12 ± 9.2
1005 (93)	2	6407 (3.81 ± 0.24) 40	1.22 ± 0.31	0.06 ± 0.1 (27)	52 (3066) 3 ± 3.2
1013 (93)	1	6511 (3.81 ± –) 41	0.91 ± –	0.02 ± – (36)	29 (5405) 5 ± –
GMB6B (95)	2	–	–	–	20 (3643) 14 ± 3.7
959 (81)	2	–	–	–	17 (2803) 7 ± 7.3

^a Mean percent viability recorded across the replicate assays.^b Median lethal concentration (LC₅₀) as viable conidia/mm²; SE = standard error.^c Spearman's coefficient of rank correlation (*r*_s) between LC₅₀ and Median lethal concentration ratio = 0.944.^d Maximum% mortality (excluding 100% values); dose (viable conidia/mm²) producing indicated mortality; C = mean% control mortality.

to include a 3-mm-diameter ventilation hole centered over each well. The lid was lined with several layers of paper toweling that formed a gasket, preventing escape of small larvae. The ventilation holes were blocked by the paper gasket, but nevertheless provided sufficient air exchange to maintain the cells at <100% relative humidity (no moisture condensation). Lids were secured with rubber bands. Larvae were monitored daily for mortality, dead larvae were removed and discarded, and fresh diet was added as needed.

Conidial viability was determined after 16–24 h by direct observation of the conidia on the YEA plates at 400× magnification. Conidia with any detectable germ tube formation were considered as viable. Samples of 100 conidia were characterized in each of three locations on the YEA plate for a total of 300 conidia. Conidial depositions from all spray applications were quantified essentially using the protocol described by Wraight et al. (1998); however, in this case, five drops of lactic acid with acid fuchsin (1 mg/ml) were placed on the treated dish surface and covered with 22-mm square coverslips. Conidia in a designated area were then counted at each of these five locations. The individual sample areas ranged from 0.025 to 1.75 mm², depending on the application rate. Adjustments in area were made generally so as to count no more than 30–

50 conidia per sample area. Final application rates were determined by adjusting numbers of conidia/mm² by the viability.

The screening bioassays were conducted over an extended period of time (January 1998–April 2000), and effort was made to ensure as random an order of testing as possible (across both fungal isolates and lepidopteran species). This protocol could not be strictly followed, however, as promising new isolates identified by Mycotech's many collaborators were added to the project at various times. Sets of screening assays conducted on a single day typically included five isolates of unknown virulence. Each set of assays also included strain GHA as a "standard". Six replicate batches of 12 larvae were treated with blank 0.01% Silwet solution as spray controls.

After completion of the screening assays, isolates were ranked within each of the principal five pest species (diamondback moth, European corn borer, corn earworm, fall armyworm and beet armyworm) by LC₅₀, median lethal concentration ratio (median LC ratio or MLCR, see Section 2.5), and the average of the LC₅₀ and MLCR ranks. These rankings provided an indication of relative virulence; however, limited replication, the large number of isolates tested, and the lack of strict randomization of assay order

Table 4Virulence of 43 isolates of *Beauveria bassiana* against European corn borer.

Isolate (% viability) ^a	Assays (n)	LC ₅₀ (log LC ₅₀ ± SE) rank ^{b,c}	Slope ± SE	Median LC ratio ± SE (rank) ^c	% Mortality (dose) C ± SE ^d
1060 (93)	3	8 (0.91 ± 0.21) 1	1.26 ± 0.18	134.4 ± 78.3 (4)	94 (120) 13 ± 2.8
1023 (91)	4	10 (1.02 ± 0.11) 2	1.12 ± 0.16	194.9 ± 88.5 (2)	92 (31) 18 ± 7.1
1014 (91)	3	12 (1.06 ± 0.30) 3	1.17 ± 0.09	74.5 ± 24.1 (7)	94 (119) 16 ± 5.0
1512 (91)	3	14 (1.15 ± 0.06) 4	1.36 ± 0.32	251.7 ± 183.7 (1)	91 (75) 13 ± 0.5
1513 (91)	3	18 (1.26 ± 0.27) 5	1.56 ± 0.48	72.0 ± 48.1 (8)	94 (171) 11 ± 2.5
1009 (81)	4	19 (1.29 ± 0.15) 6	1.18 ± 0.15	149.1 ± 110.0 (3)	98 (415) 21 ± 6.2
1020 (93)	3	21 (1.31 ± 0.32) 7	1.22 ± 0.17	25.4 ± 5.1 (17)	94 (246) 18 ± 4.9
GMB6B (92)	3	23 (1.36 ± 0.21) 8	1.05 ± 0.27	62.2 ± 23.7 (10)	95 (791) 12 ± 3.6
1067 (90)	3	23 (1.37 ± 0.25) 9	1.29 ± 0.33	50.9 ± 28.8 (12)	94 (220) 16 ± 1.4
1019 (90)	3	24 (1.37 ± 0.43) 10	1.26 ± 0.11	24.9 ± 10.5 (18)	91 (108) 18 ± 4.9
1022 (92)	4	25 (1.39 ± 0.16) 11	1.17 ± 0.15	90.5 ± 37.8 (6)	92 (40) 19 ± 6.8
1015 (77)	4	30 (1.47 ± 0.24) 12	1.03 ± 0.06	131.0 ± 86.4 (5)	97 (203) 21 ± 6.0
1013 (97)	2	31 (1.50 ± 0.72) 13	0.76 ± 0.29	55.2 ± 25.6 (11)	86 (826) 16 ± 1.9
1012 (96)	2	35 (1.55 ± 0.04) 14	1.41 ± 0.02	29.1 ± 9.1 (15)	98 (817) 22 ± 3.4
1008 (73)	2	43 (1.63 ± 0.07) 15	1.24 ± 0.13	34.8 ± 25.1 (13)	95 (1038) 16 ± 2.5
1005 (89)	2	44 (1.64 ± 0.20) 16	1.06 ± 0.06	63.8 ± 15.3 (9)	97 (945) 11 ± 2.0
1835 (94)	2	82 (1.92 ± 0.27) 17	1.56 ± 0.01	11.2 ± 7.1 (20)	36 (27) 10 ± 3.9
1200 (93)	4	98 (1.99 ± 0.30) 18	1.63 ± 0.43	27.1 ± 13.4 (16)	92 (128) 19 ± 7.1
1003 (93)	3	157 (2.20 ± 0.01) 19	1.15 ± 0.14	31.4 ± 13.04 (14)	84 (1059) 17 ± 4.6
1966 (88)	2	568 (2.75 ± 0.34) 20	1.15 ± 0.42	1.8 ± 1.3 (29)	66 (2605) 10 ± 3.9
1055 (93)	1	588 (2.77 ± –) 21	0.93 ± –	2.5 ± – (26)	88 (5130) 14 ± –
1082 (90)	2	698 (2.84 ± 0.36) 22	1.32 ± 0.28	7.5 ± 2.3 (22)	62 (626) 12 ± 5.2
SF20 (95)	2	702 (2.85 ± 0.04) 23	1.22 ± 0.22	5.8 ± 0.5 (23)	63 (1030) 9 ± 0.5
SA8 (93)	3	762 (2.88 ± 0.20) 24	0.72 ± 0.07	1.73 ± 0.9 (30)	78 (2771) 18 ± 8.4
1059 (97)	2	846 (2.93 ± 0.15) 25	1.77 ± 0.25	1.15 ± 0.3 (32)	84 (1751) 33 ± 1.6
1153 (93)	2	857 (2.93 ± 0.15) 26	1.36 ± 0.40	4.6 ± 2.5 (24)	80 (2002) 13 ± 0.2
1075 (91)	2	859 (2.93 ± 0.03) 27	1.66 ± 0.39	0.90 ± 0.4 (34)	62 (1719) 21 ± 3.1
2559 (94)	2	892 (2.95 ± 0.30) 28	1.27 ± 0.42	2.4 ± 0.7 (27)	90 (1824) 11 ± 5.8
2638 (88)	2	986 (2.99 ± 0.24) 29	1.68 ± 1.27	3.1 ± 2.3 (25)	88 (1360) 11 ± 5.8
1083 (93)	2	1134 (3.05 ± 0.07) 30	1.12 ± 0.49	11.9 ± 4.2 (19)	91 (4344) 14 ± 7.9
1084 (92)	2	1614 (3.21 ± 0.06) 31	1.16 ± 0.53	8.3 ± 2.8 (21)	64 (2025) 14 ± 7.9
GHA (93)	33	1668 (3.22 ± 0.08) 32	1.20 ± 0.08	–	98 (5468) 16 ± 1.5
SE6 (94)	2	2200 (3.34 ± 0.08) 33	0.95 ± 0.05	2.0 ± 0.7 (28)	34 (1587) 9 ± 0.5
1086 (93)	2	2232 (3.35 ± 0.09) 34	0.82 ± 0.02	0.55 ± 0.1 (37)	39 (1943) 7 ± 2.2
3288 (95)	2	2433 (3.39 ± 0.34) 35	1.29 ± 0.48	0.90 ± 0.3 (34)	49 (1691) 9 ± 3.6
1077 (96)	2	2830 (3.45 ± 0.11) 36	0.78 ± 0.25	0.55 ± 0.2 (37)	51 (2305) 16 ± 1.1
SC19 (96)	2	3191 (3.50 ± 0.46) 37	1.38 ± 0.67	0.70 ± 0.4 (36)	57 (1301) 15 ± 9.1
3286 (90)	1	3255 (3.51 ± –) 38	1.11 ± –	0.93 ± – (33)	24 (753) 9 ± –
1076 (89)	2	4562 (3.66 ± 0.05) 39	1.03 ± 0.57	1.70 ± 1.3 (31)	37 (940) 12 ± 5.2
SC15 (95)	3	4995 (3.70 ± 0.23) 40	1.11 ± 0.46	0.50 ± 0.1 (39)	38 (1141) 16 ± 9.1
2945 (84)	2	–	–	–	28 (982) 11 ± 5.7
1085 (95)	2	–	–	–	10 (1916) 14 ± 8.7
959 (83)	2	–	–	–	2 (2319) 13 ± 0.2

^a Mean percent viability recorded across the replicate assays.^b Median lethal concentration (LC₅₀) as viable conidia/mm²; SE = standard error.^c Spearman's coefficient of rank correlation (r_s) between LC₅₀ and Median lethal concentration ratio = 0.899.^d Maximum% mortality (excluding 100% values); dose (viable conidia/mm²) producing indicated mortality; C = mean% control mortality.

precluded rigorous analysis for isolate separation. Therefore, the top-ranked isolates against each pest species were compared directly in replicated tests (see following section).

To identify isolates with greatest virulence against the broadest range of lepidopteran species, rank scores assigned to the 23 most virulent isolates were analyzed independently. The mean LC₅₀ and mean MLCR ranks across all pest species were calculated and compared. The grand mean ranks from these two ranking systems were also examined.

2.4. Direct comparisons of virulent isolates within pest species

The four or five isolates ranking highest in virulence against each pest (based on a ranking of LC₅₀s) were compared directly in replicated tests. Each test comprised 2–3 independent replicate assays conducted within a 12-h (1-day) period. Each of these tests also included the standard strain GHA, and after the exceptional virulence of isolate 1200 was discovered, this isolate was included in all comparisons. Assays were conducted under a randomized complete block design with the different tests (days) serving as the blocks. Assay protocols were identical to the screening assays,

except that each isolate was tested at five conidial concentrations prepared from serial threefold dilutions, a different dilution series was prepared for each replicate assay, and in each replicate assay a single batch of 12 larvae was treated per dose (60 larvae/assay).

During the screening process, it became evident that various groups of isolates produced similar responses in all test insects. Subsequent RFLP analysis of the corn isolates (Mycotech isolates 1003–1023 indicated no differences among these fungi (Mycotech, unpublished data). Preliminary PCR analysis also indicated no differences between isolates 1512 and 1513 or between isolates SC15 and SC19 and suggested a close relationship among isolates 1082, 1083, and 1084 (L. Castrillo, unpublished data). These groupings were taken into consideration in ranking the top isolates; when strains from these groups appeared in the top five rankings, they were ranked together and listed in order of appearance. Also, when two or more strains from a group of closely related isolates fell in the top rankings, usually only one was selected for the comparative virulence assays. As a check of assay precision, however, two isolates with expected similar efficacy were included in the comparative virulence assays with two pest species (isolates 1082 and 1084 were tested against diamondback moth and isolates 1009

Table 5Virulence of 43 isolates of *Beauveria bassiana* against corn earworm.

Isolate (% viability) ^a	Assays (n)	LC ₅₀ (log LC ₅₀ ± SE) rank ^{b,c}	Slope ± SE	Median LC ratio ± SE (rank) ^c	% Mortality (dose) C ± SE ^d
3288 (93)	2	2 (0.26 ± 0.64) 1	1.50 ± 0.82	5.60 ± 5.2 (2)	97 (27) 3 ± 2.8
SC15 (92)	3	3 (0.47 ± 0.40) 2	1.72 ± 0.28	3.06 ± 1.8 (3)	89 (131) 3 ± 1.8
1200 (90)	3	4 (0.59 ± 0.06) 3	2.65 ± 0.70	7.95 ± 7.1 (1)	91 (16) 0 ± 0
1075 (87)	2	4 (0.60 ± 0.30) 4	1.58 ± 0.52	2.81 ± 1.7 (4)	75 (11) 5 ± 1.2
1153 (88)	2	5 (0.72 ± 0.42) 5	1.29 ± 0.23	0.69 ± 0.3 (13)	94 (376) 11 ± 7.9
1083 (95)	3	6 (0.76 ± 0.22) 6	1.86 ± 0.16	1.70 ± 0.6 (6)	83 (5) 5 ± 0.7
1005 (91)	3	6 (0.79 ± 0.46) 7	1.06 ± 0.01	1.17 ± 0.6 (9)	91 (54) 8 ± 5.4
2638 (89)	2	8 (0.90 ± 1.20) 8	1.41 ± 0.79	2.02 ± 1.6 (5)	97 (1012) 4 ± 4.2
1003 (93)	3	8 (0.91 ± 0.26) 9	1.16 ± 0.11	0.59 ± 0.1 (15)	92 (32) 8 ± 5.4
GHA (92)	35	9 (0.95 ± 0.10) 10	1.22 ± 0.09	–	95 (142) 5 ± 0.8
1077 (90)	2	9 (0.95 ± 0.01) 10	1.53 ± 0.56	1.08 ± 0.03 (10)	58 (10) 5 ± 1.2
SA8 (94)	3	10 (0.98 ± 0.14) 12	1.07 ± 0.14	0.90 ± 0.4 (11)	96 (382) 6 ± 0.6
SC19 (94)	2	13 (1.12 ± 0.08) 13	1.53 ± 0.10	1.21 ± 1.0 (7)	94 (255) 3 ± 3.1
1055 (88)	2	14 (1.15 ± 0.55) 14	1.35 ± 0.13	0.81 ± 0.1 (12)	97 (92) 0 ± 0
1084 (91)	2	16 (1.21 ± 0.31) 15	1.70 ± 0.40	1.19 ± 0.01 (8)	73 (16) 4 ± 0.2
SF20 (94)	2	18 (1.25 ± 0.02) 16	1.36 ± 0.28	0.37 ± 0.2 (20)	97 (145) 5 ± 0.4
2945 (88)	2	20 (1.30 ± 1.00) 17	0.67 ± 0.05	0.51 ± 0.5 (16)	75 (57) 4 ± 4.2
1013 (93)	2	24 (1.39 ± 0.13) 18	1.01 ± 0.22	0.11 ± 0.02 (31)	92 (148) 7 ± 1.9
1020 (94)	2	32 (1.51 ± 0.61) 19	1.34 ± 0.61	0.24 ± 0.0 (24)	92 (973) 9 ± 1.2
1512 (90)	2	34 (1.53 ± 0.33) 20	1.35 ± 0.16	0.11 ± 0.03 (31)	94 (819) 11 ± 7.9
1086 (95)	2	37 (1.57 ± 0.30) 21	0.68 ± 0.25	0.13 ± 0.03 (30)	85 (1971) 5 ± 0.4
1015 (95)	2	37 (1.57 ± 0.53) 21	0.86 ± 0.59	0.37 ± 0.2 (20)	92 (613) 5 ± 3.2
1067 (95)	2	46 (1.66 ± 0.40) 23	1.77 ± 0.70	0.28 ± 0.2 (23)	94 (154) 8 ± 1.5
1014 (94)	2	47 (1.68 ± 0.01) 24	0.82 ± 0.03	0.06 ± 0.03 (34)	91 (186) 7 ± 1.9
1082 (97)	2	49 (1.69 ± 0.17) 25	1.63 ± 0.47	0.41 ± 0.1 (19)	91 (102) 4 ± 0.2
1059 (92)	2	50 (1.70 ± 0.22) 26	1.53 ± 0.31	0.34 ± 0.2 (22)	94 (229) 0 ± 0
SE6 (92)	2	56 (1.75 ± 0.35) 27	1.39 ± 0.25	0.17 ± 0.02 (29)	67 (55) 3 ± 3.1
1966 (86)	2	62 (1.79 ± 0.71) 28	1.26 ± 0.29	0.21 ± 0.02 (25)	57 (23) 1 ± 1.4
1022 (96)	2	66 (1.82 ± 0.41) 29	1.66 ± 1.25	0.62 ± 0.6 (14)	80 (318) 5 ± 3.0
1835 (88)	2	73 (1.86 ± 0.54) 30	1.32 ± 0.36	0.18 ± 0.1 (28)	70 (45) 1 ± 1.4
GMB6B (90)	2	91 (1.96 ± 0.13) 31	1.33 ± 0.41	0.05 ± 0.01 (37)	83 (286) 5 ± 0.4
1085 (97)	2	95 (1.98 ± 0.05) 32	1.90 ± 0.98	0.06 ± 0.02 (34)	94 (1197) 5 ± 0.4
1076 (90)	2	117 (2.07 ± 0.55) 33	0.76 ± 0.22	0.10 ± 0.1 (33)	69 (1818) 8 ± 1.5
3286 (92)	2	130 (2.11 ± 0.51) 34	0.89 ± 0.27	0.03 ± 0.02 (39)	81 (309) 4 ± 4.2
1019 (89)	2	134 (2.13 ± 0.24) 35	0.62 ± 0.11	0.49 ± 0.5 (17)	77 (938) 6 ± 4.2
2559 (92)	2	163 (2.21 ± 0.15) 36	1.65 ± 0.54	0.47 ± 0.01 (18)	89 (740) 0 ± 0
1012 (94)	2	175 (2.24 ± 0.34) 37	0.85 ± 0.06	0.02 ± 0.01 (43)	94 (839) 7 ± 1.9
1009 (92)	2	198 (2.30 ± 0.64) 38	1.17 ± 0.09	0.19 ± 0.04 (26)	92 (681) 10 ± 3.6
1008 (96)	2	199 (2.30 ± 0.57) 39	1.19 ± 0.11	0.19 ± 0.01 (26)	97 (736) 10 ± 3.6
1023 (95)	2	280 (2.45 ± 0.10) 40	0.83 ± 0.07	0.06 ± 0.03 (34)	60 (323) 5 ± 3.0
1513 (88)	2	450 (2.65 ± 0.70) 41	1.03 ± 0.05	0.03 ± 0.0 (40)	91 (1035) 1 ± 1.4
959 (83)	2	523 (2.72 ± 0.54) 42	1.09 ± 0.12	0.03 ± 0.01 (40)	67 (705) 1 ± 1.4
1060 (91)	2	829 (2.92 ± 0.17) 43	0.93 ± 0.29	0.05 ± 0.04 (37)	67 (1454) 0 ± 0

^a Mean percent viability recorded across the replicate assays.^b Median lethal concentration (LC₅₀) as viable conidia/mm²; SE = standard error.^c Spearman's coefficient of rank correlation (r_s) between LC₅₀ and Median lethal concentration ratio = 0.830.^d Maximum% mortality (excluding 100% values); dose (viable conidia/mm²) producing indicated mortality; C = mean% control mortality.

and 1023 were tested against European corn borer). In the following presentations, isolate groups are indicated and abbreviated by use of slashes; e.g., the 1082, 1083, 1084 group is designated 1082/83/84. These particular isolates are also referred to as the 1080 isolates. Isolates 1003–1023, all isolated from corn plants (corn endophytes) are referred to as the corn isolates.

2.5. Statistical analyses

LC₅₀s were estimated using the PoloPlus program (LeOra Software, 2002–2003). In analyzing the 3-dose screening assays, responses from the pseudoreplicated batches of insects exposed to each dose were pooled for probit analysis. In contrast, each replicate of the 5-dose comparative virulence assays was analyzed independently (producing replicate estimates of LC₅₀). Results were adjusted for control mortality by the PoloPlus program.

The multiple 3-dose screening bioassays conducted on a given day were analyzed independently (intercepts and slopes unconstrained) and in combination (slopes constrained and both slopes and intercepts constrained) via a single run of the PoloPlus program. We used the slope and LC₅₀ estimates from the indepen-

dently generated regression lines whenever possible; however, these assays, being based on few doses, did not always provide reliable estimates of slope. When this problem occurred, the chi-square test for parallelism was examined, and if no significant differences among slopes were indicated, parameter estimates were obtained from the probit line fit to the common slope (intercept unconstrained). If parallelism was rejected, the problem assay was discarded.

In addition to the LC₅₀ for each assay, an additional variable, the median LC ratio, was calculated to serve as a potentially more stable measure of virulence (Robertson and Preisler, 1992; Robertson et al., 2007). This ratio was calculated by dividing the LC₅₀ of the selected standard (commercial strain GHA) by the LC₅₀ of the unknown (LC₅₀s expressed in linear scale; i.e., as conidia/mm²).

ANOVAs, nonparametric tests, and correlation analyses were conducted using the JMP statistical software (SAS Institute, 2000). Assays were blocked over time; block interactions were not tested due to potential restriction error (Sokal and Rohlf, 1995). LC₅₀ values are derived as the ratios of the variables from the linear equations of log dose-probit mortality lines (intercept/slope), and ratio data are generally not normally distributed

Table 6Virulence of 43 isolates of *Beauveria bassiana* against fall armyworm.

Isolate (% viability) ^a	Assays (n)	LC ₅₀ (log LC ₅₀ ± SE) rank ^{b,c}	Slope ± SE	Median LC ratio ± SE (rank) ^c	% Mortality (dose) C ± SE ^d
1200 (92)	3	11 (1.05 ± 0.13) 1	1.79 ± 0.51	163.2 ± 123.6 (1)	97 (67) 4 ± 3.8
SC15 (92)	4	170 (2.23 ± 0.21) 2	1.40 ± 0.33	8.78 ± 3.5 (4)	83 (878) 12 ± 5.0
SC19 (94)	3	235 (2.37 ± 0.15) 3	1.64 ± 0.29	6.59 ± 4.2 (7)	89 (1131) 13 ± 7.5
1083 (96)	3	249 (2.40 ± 0.06) 4	1.38 ± 0.13	13.62 ± 4.2 (2)	98 (2703) 14 ± 1.6
1513 (93)	4	280 (2.45 ± 0.13) 5	1.33 ± 0.31	5.30 ± 1.7 (8)	83 (1254) 1 ± 0.5
1153 (91)	4	289 (2.46 ± 0.20) 6	1.65 ± 0.22	3.62 ± 1.9 (12)	81 (495) 4 ± 2.1
1084 (95)	3	308 (2.49 ± 0.15) 7	1.22 ± 0.18	10.84 ± 2.3 (3)	95 (2555) 14 ± 1.6
1512 (96)	2	332 (2.52 ± 0.13) 8	1.12 ± 0.18	3.20 ± 1.1 (16)	67 (975) 4 ± 3.8
1019 (95)	2	350 (2.54 ± 0.25) 9	1.31 ± 0.16	1.57 ± 0.5 (21)	93 (2942) 13 ± 8.5
1022 (94)	2	381 (2.58 ± 0.01) 10	2.47 ± 0.17	1.51 ± 0.3 (22)	93 (1514) 28 ± 0.7
1082 (91)	3	405 (2.61 ± 0.21) 11	2.21 ± 0.98	7.59 ± 2.6 (6)	98 (1877) 16 ± 2.9
1009 (73)	3	415 (2.62 ± 0.22) 12	1.76 ± 0.43	3.59 ± 0.3 (14)	94 (3436) 18 ± 2.1
1003 (93)	2	440 (2.64 ± 0.26) 13	1.45 ± 0.10	2.67 ± 0.4 (18)	85 (1206) 20 ± 1.3
1059 (95)	3	493 (2.69 ± 0.10) 14	1.77 ± 0.28	3.91 ± 1.3 (10)	83 (1330) 17 ± 7.6
2559 (92)	1	546 (2.74 ± –) 15	0.95 ± –	3.26 ± – (15)	67 (897) 1 ± –
SA8 (95)	2	551 (2.74 ± 0.07) 16	1.41 ± 0.17	1.40 ± 0.4 (24)	95 (465) 19 ± 8.7
1075 (95)	2	578 (2.76 ± 0.14) 17	1.17 ± 0.39	0.99 ± 0.5 (30)	96 (2981) 21 ± 5.4
2638 (84)	5	582 (2.77 ± 0.17) 18	1.38 ± 0.24	7.92 ± 2.9 (5)	97 (3701) 4 ± 2.0
1012 (95)	2	603 (2.78 ± 0.32) 19	1.75 ± 0.60	1.33 ± 0.3 (26)	96 (1432) 24 ± 2.9
1014 (95)	2	609 (2.78 ± 0.35) 20	1.42 ± 0.27	1.35 ± 0.4 (25)	97 (6676) 25 ± 2.9
SF20 (94)	4	666 (2.82 ± 0.28) 21	1.04 ± 0.41	4.10 ± 1.9 (9)	67 (2827) 13 ± 4.5
1060 (96)	3	675 (2.83 ± 0.57) 22	1.32 ± 0.17	1.31 ± 0.5 (27)	92 (1699) 16 ± 6.6
SE6 (94)	3	677 (2.83 ± 0.27) 23	0.96 ± 0.15	2.93 ± 0.4 (17)	91 (5144) 18 ± 5.1
1008 (78)	2	695 (2.84 ± 0.47) 24	1.86 ± 0.70	1.45 ± 0.6 (23)	94 (6566) 18 ± 2.1
GMB6B (94)	3	717 (2.86 ± 0.32) 25	1.16 ± 0.20	3.73 ± 0.7 (11)	82 (1298) 15 ± 2.2
1005 (94)	2	767 (2.88 ± 0.58) 26	1.39 ± 0.28	1.74 ± 0.6 (20)	97 (5966) 26 ± 1.0
1966 (90)	2	844 (2.93 ± 0.92) 27	1.17 ± 0.26	1.25 ± 0.7 (28)	87 (923) 6 ± 3.7
1086 (95)	2	953 (2.98 ± 0.06) 28	1.32 ± 0.03	0.45 ± 0.1 (35)	83 (3560) 17 ± 6.3
1067 (97)	2	1151 (3.06 ± 0.35) 29	1.66 ± 0.67	1.12 ± 0.2 (29)	85 (1255) 21 ± 6.5
1055 (93)	3	1179 (3.07 ± 0.36) 30	1.22 ± 0.26	3.59 ± 1.1 (13)	95 (4095) 16 ± 2.9
GHA (94)	33	1213 (3.08 ± 0.08) 31	1.38 ± 0.07	–	95 (4234) 13 ± 1.8
1020 (94)	3	1257 (3.10 ± 0.42) 32	1.46 ± 0.13	0.88 ± 0.3 (33)	98 (2594) 16 ± 6.1
1015 (89)	2	1475 (3.17 ± 0.28) 33	1.32 ± 0.17	2.11 ± 0.5 (19)	72 (3404) 21 ± 0.4
1023 (95)	2	1506 (3.18 ± 0.28) 34	1.26 ± 0.36	0.96 ± 0.1 (31)	76 (5669) 23 ± 5.4
1013 (94)	2	1646 (3.22 ± 0.35) 35	1.70 ± 0.41	0.69 ± 0.19 (34)	84 (3350) 25 ± 1.6
1076 (92)	2	1807 (3.26 ± 0.39) 36	1.23 ± 0.45	0.31 ± 0.04 (36)	74 (3662) 22 ± 3.9
1085 (96)	2	2118 (3.33 ± 0.05) 37	1.96 ± 0.67	0.19 ± 0.01 (38)	72 (6995) 17 ± 6.2
3288 (90)	1	3462 (3.54 ± –) 38	1.51 ± –	0.21 ± – (37)	69 (4325) 0 ± –
1077 (94)	2	4092 (3.61 ± 0.23) 39	1.30 ± 0.31	0.92 ± 0.5 (32)	61 (5385) 16 ± 1.8
2945 (86)	2	– (–)	–	–	15 (1067) 1 ± 1.1
1835 (89)	2	– (–)	–	–	3 (498) 5 ± 4.8
3286 (93)	2	– (–)	–	–	3 (1213) 0 ± 0
959 (92)	2	– (–)	–	–	3 (3440) 5 ± 4.8

^a Mean percent viability recorded across the replicate assays.^b Median lethal concentration (LC₅₀) as viable conidia/mm²; SE = standard error.^c Spearman's coefficient of rank correlation (r_s) between LC₅₀ and Median lethal concentration ratio = 0.784.^d Maximum% mortality (excluding 100% values); dose (viable conidia/mm²) producing indicated mortality; C = mean% control mortality.

(Anderson and Lydic, 1977; Sokal and Rohlf, 1995). The large numbers of screening assays of the selected standard strain (GHA) conducted against five host species provided an unique opportunity for detailed investigation of the nature of the distributions of the variables obtained from the probit regressions. Knowledge of these distributions is essential for rigorous statistical analysis, most importantly with respect to selection of parametric versus non-parametric methods. Estimates of slope, intercept, and log LC₅₀ from each replicate assay of strain GHA conducted with each insect host were submitted to the Shapiro–Wilk test for normality provided by the JMP program. This test was designed to account for all possible alternatives to normality and is highly recommended as being especially sensitive to asymmetry and heavy tailedness (Shapiro and Wilk, 1965; D'Agostino et al., 1990; Thode, 2002). Normal quantile plots of the data and plots of ANOVA residuals were also examined for systematic departures from the model.

LC₅₀ data from the comparative virulence assays of the top-ranked isolates were analyzed via two methods: (1) standard parametric ANOVA followed by the Tukey HSD test for all pairwise comparisons and (2) the nonparametric Kruskal–Wallis test on data adjusted for block effects by the JMP program. The latter is

an aligned-rank method with greater power than the standard Friedman test for randomized complete blocks (O'Gorman, 2001). There are numerous options for multiple comparisons following the Kruskal–Wallis or Friedman tests (see Sheskin, 2007). We opted to conduct a Wilcoxon two-sample test for each pairwise comparison (also using aligned ranks) and determine significant differences via the sequential Bonferroni method of Holm (1979) (see also Sokal and Rohlf, 1995; Shaffer, 1995). This approach is one of the alternatives recommended by Sheskin (2007) for all pairwise comparisons following the Friedman test. Results from the two procedures were compared, and recommendations are offered for routine analysis of replicated LC₅₀ data. Log dose-probit regression slopes were analyzed using the parametric ANOVA/Tukey HSD tests.

3. Results

3.1. Screening bioassays

Results of the screening assays, including isolate rankings by LC₅₀ and MLCR are presented in Tables 3–7. The top isolates based

Table 7Virulence of 29 isolates of *Beauveria bassiana* against beet armyworm.

Isolate (% viability) ^a	Assays (n)	LC ₅₀ (log LC ₅₀ ± SE) rank ^{b,c}	Slope ± SE	Median LC ratio ± SE (rank) ^c	% Mortality (dose) C ± SE ^d
1200 (96)	3	5 (0.71 ± 0.14) 1	2.70 ± 0.23	21.94 ± 5.5 (1)	97 (29) 1 ± 0.7
GMB6B (92)	2	10 (1.00 ± 0.04) 2	2.55 ± 0.23	5.81 ± 3.3 (10)	83 (22) 5 ± 4.7
1023 (95)	2	13 (1.11 ± 0.27) 3	2.05 ± 0.05	7.19 ± 4.1 (5)	81 (64) 7 ± 1.1
1084 (93)	2	13 (1.11 ± 0.03) 3	1.77 ± 0.42	6.62 ± 3.3 (7)	97 (89) 1 ± 1.0
1083 (92)	2	13 (1.13 ± 0.05) 5	2.11 ± 0.07	7.04 ± 4.5 (6)	92 (68) 1 ± 1.0
SE6 (92)	2	14 (1.14 ± 0.29) 6	1.39 ± 0.11	6.52 ± 4.8 (8)	92 (209) 5 ± 3.6
SC19 (94)	2	14 (1.14 ± 0.24) 6	1.55 ± 0.06	4.55 ± 1.2 (12)	89 (137) 5 ± 3.6
1082 (94)	3	14 (1.16 ± 0.29) 8	1.36 ± 0.27	8.04 ± 5.9 (3)	94 (217) 3 ± 2.6
1009 (86)	3	16 (1.19 ± 0.23) 9	2.11 ± 0.08	8.12 ± 5.8 (2)	94 (94) 3 ± 1.9
1059 (94)	2	16 (1.21 ± 0.07) 10	2.17 ± 0.16	2.44 ± 1.6 (18)	94 (125) 5 ± 4.7
1022 (96)	2	18 (1.25 ± 0.07) 11	1.75 ± 0.26	3.31 ± 0.3 (15)	94 (101) 2 ± 2.1
1513 (95)	2	21 (1.32 ± 0.02) 12	2.34 ± 0.58	4.82 ± 1.0 (11)	97 (220) 5 ± 1.2
2638 (92)	2	23 (1.36 ± 0.08) 13	1.66 ± 0.24	2.57 ± 0.6 (17)	94 (176) 2 ± 2.1
SC15 (88)	3	25 (1.40 ± 0.31) 14	1.71 ± 0.24	7.70 ± 6.04 (4)	92 (480) 4 ± 2.6
1060 (87)	2	25 (1.40 ± 0.32) 14	2.22 ± 0.04	6.13 ± 5.4 (9)	83 (151) 1 ± 1.0
SA8 (93)	3	27 (1.43 ± 0.22) 16	1.35 ± 0.22	4.46 ± 1.84 (13)	86 (106) 6 ± 2.0
1966 (92)	2	28 (1.45 ± 0.02) 17	1.80 ± 0.09	2.06 ± 0.2 (21)	83 (122) 2 ± 2.1
2559 (96)	2	29 (1.47 ± 0.21) 18	2.02 ± 0.13	2.31 ± 1.2 (19)	94 (93) 2 ± 2.1
1512 (91)	2	32 (1.50 ± 0.01) 19	2.60 ± 0.32	3.16 ± 0.4 (16)	97 (176) 5 ± 1.2
SF20 (91)	2	33 (1.52 ± 0.12) 20	2.11 ± 0.62	1.86 ± 0.02 (22)	94 (160) 5 ± 3.6
3288 (92)	3	34 (1.54 ± 0.11) 21	2.57 ± 0.23	2.11 ± 0.6 (20)	89 (79) 1 ± 0.7
1055 (87)	1	35 (1.54 ± –) 22	2.32 ± –	0.81 ± – (25)	93 (189) 9 ± –
1153 (91)	3	41 (1.61 ± 0.08) 23	2.44 ± 0.25	1.80 ± 0.4 (23)	80 (82) 4 ± 1.9
1075 (94)	2	56 (1.75 ± 0.06) 24	1.71 ± 0.31	3.84 ± 2.7 (14)	92 (118) 5 ± 3.4
GHA (92)	15	67 (1.83 ± 0.05) 25	1.86 ± 0.17	–	92 (167) 3 ±
2945 (86)	2	72 (1.86 ± 0.07) 26	1.66 ± 0.63	1.32 ± 0.02 (24)	83 (141) 1 ± 1.0
1835 (90)	2	341 (2.53 ± 0.89) 27	1.17 ± 0.73	0.71 ± 0.7 (26)	72 (110) 1 ± 1.0
3286 (90)	2	423 (2.63 ± 0.07) 28	1.85 ± 0.45	0.23 ± 0.01 (27)	80 (1600) 1 ± 1.0
959 (92)	2	–	–	–	6 (2391) 5 ± 1.2

^a Mean percent viability recorded across the replicate assays.^b Median lethal concentration (LC₅₀) as viable conidia/mm²; SE = standard error.^c Spearman's coefficient of rank correlation (r_s) between LC₅₀ and Median lethal concentration ratio = 0.825.^d Maximum% mortality (excluding 100% values); dose (viable conidia/mm²) producing indicated mortality; C = mean% control mortality.

on each of these rankings and the grand mean of these rankings are presented in Table 8. Mortality among spray controls was erratically high in assays of fall armyworm and European corn borer; the average rate exceeding 20% in assays of 10 of the 43 isolates against armyworm and in assays of five of the 43 isolates against corn borer. However, most standard errors associated with these values were small (mean CV = 15.1%), and the grand mean control mortality for the respective species was only 14.0% and 14.7%, respectively. It appeared that these insects were sensitive to *Aspergillus* contamination of the diet that became a problem at times due to the absence of antimicrobial (fungicidal) ingredients in the larval diets.

A strong inverse relationship between the average rate of control mortality and susceptibility of the various pests to fungal infection is suggested by the data (the most susceptible pests exhibiting lowest control mortalities); however, this is not substantiated by analysis. ANOVA of the LC₅₀/control mortality data (log/arcsine transformed) from 24 isolates that were successfully assayed against DBM, ECB, CEW, FAW, and BAW reveals that while LC₅₀ was highly dependent upon pest species ($F_{4, 114} = 12.5$, $P < 0.0001$), control mortality was an insignificant covariate ($F_{1, 114} = 2.73$, $P = 0.10$). Addition of control mortality to the model increased r^2 only slightly, from 0.395 to 0.413. There was no pest species × control mortality interaction ($F_{4, 114} = 0.49$, $P = 0.75$). Results were similar from ANOVA of an alternative set of data from the 37 isolates assayed against DBM, ECB, CEW, and FAW (pest species $F_{3, 140} = 20.7$, $P < 0.0001$; control mortality $F_{1, 140} = 3.05$, $P = 0.083$; pest × control mortality $F_{3, 140} = 1.89$, $P = 0.13$). Addition of control mortality as a covariate increased r^2 by <3 percentage points (0.315–0.344). It is also interesting to note that within those pests exhibiting highest control mortalities (and greatest ranges of control mortalities), correlations between the mean control mortalities and LC₅₀s associated with each isolate were not consistently positive or negative, and none was statistically significant

(diamondback moth: $r = -0.101$, $P = 0.53$, $n = 41$; European corn borer: $r = -0.292$, $P = 0.068$, $n = 40$; fall armyworm: $r = 0.229$, $P = 0.16$, $n = 39$). In addition, the standard strain GHA was tested in all assays with these pests, and there were no significant correlations between control mortalities and LC₅₀s of this pathogen (diamondback moth: $r = 0.276$, $P = 0.15$, $n = 28$; European corn borer: $r = 0.048$, $P = 0.78$, $n = 33$; fall armyworm: $r = 0.077$, $P = 0.67$, $n = 33$). This evidence of negligible correlation between host susceptibility to *B. bassiana* infection and control mortality supports our acceptance of bioassays with a broad range of control mortalities and use of control corrections (via the POLO program) for estimating virulence. It is noteworthy in this regard that because larvae were treated and held for 24 h in clean Petri dishes and then transferred individually to clean holding chambers, the above-mentioned mold contamination did not become problematic until after most *B. bassiana* infections were well established.

Ranking of isolates based on LC₅₀ and MLCR produced similar results (Table 8). Among 51 isolates or isolate groups ranked among the top five (the top five isolates ranked by LC₅₀ and MLCR for the five pest species), only seven (14%) were dissimilar (marked by bold print in Table 8). The most pronounced discrepancy occurred with beet armyworm; isolate GMB6B was ranked second by LC₅₀ but was not included among the top five by MLCR. The close relationship between LC₅₀ and MLCR as measures of virulence was reflected in high correlations between these variables. Values of the Spearman rank correlation coefficients (r_s) for the five insects ranged from 0.784 to 0.944 (mean 0.856) (Tables 3–7).

The LC₅₀ and MLCR rankings demonstrated that isolate 1200 had an exceptional level of virulence and an unusually broad lepidopteran host range. It was ranked among the top five isolates against four of five pest species. When the LC₅₀ and MLCR rankings were averaged, this isolate ranked first against diamondback moth, fall armyworm, and beet armyworm and second against corn

Table 8Highest ranked isolates of *Beauveria bassiana* against five Lepidoptera species.

Pest species	Isolates ranked by		
	LC ₅₀ ^a	Median LC ratio (MLCR) ^a	Mean of LC ₅₀ and MLCR ranks ^b
Diamondback moth	1 – 1200	1 – 1084/82/83	1 – 1200
	2 – 1966	2 – 1200	2 – 1084/82
	3 – 1084/82	3 – 1966	3 – 1966
	4 – 3288	4 – 3288	4 – 3288
	5 – 1075	5 – 3286	5 – 1153
Fall armyworm	1 – 1200	1 – 1200	1 – 1200
	2 – SC15/19	2 – 1083/84/82	2 – SC15/19
	3 – 1083	3 – SC15/19	2 – 1083/84/82
	4 – 1513	4 – 2638	4 – 1513
	5 – 1153	5 – 1513	5 – 1153
European corn borer	1 – 1060	1 – 1512/13	1 – 1023/09/14/15/22
	2 – 1023/14/09/20	2 – 1023/09/15/22/14/05/13	2 – 1060
	3 – 1512/13	3 – 1060	3 – 1512/13
	4 – GMB6B	4 – GMB6B	4 – GMB6B
	5 – 1067	5 – 1067	5 – 1067
Corn earworm	1 – 3288	1 – 1200	1 – 3288
	2 – SC15	2 – 3288	2 – 1200
	3 – 1200	3 – SC15	3 – SC15
	4 – 1075	4 – 1075	4 – 1075
	5 – 1153	5 – 2638	5 – 1083
Beet armyworm	1 – 1200	1 – 1200	1 – 1200
	2 – GMB6B	2 – 1009/23	2 – 1023/09
	3 – 1023	3 – 1082/83/84	3 – 1084/83/82
	3 – 1084/83	4 – SC15	4 – GMB6B
	4 – SE6	5 – SE6	5 – SE6
	4 – SC19		

^a Isolates in bold type were ranked in the top 5 by only one of the alternative ranking systems (LC₅₀ versus MLCR).^b Isolates ranked by the mean of the ranks based on LC₅₀ and MLCR (see Tables 3–7); groups of closely related (genetically similar) isolates are assigned the rank of the highest ranking member of the group, and isolates within a group are listed in decreasing rank order.

earworm (Table 8). Isolate 1200 ranked just 18th against European corn borer, but was nevertheless virulent against this pest, with an LC₅₀ <100 conidia/mm² (Table 4). Combined traits of high virulence and broad host range were indicated in only two other isolate groups. Isolates from the 1080 and SC groups were ranked among the top five against three of the five pests (Table 8).

3.2. Rankings across Lepidoptera species

The mean LC₅₀ ranks and mean MLCR ranks of the 23 most virulent isolates calculated across all pest species are presented in Table 9. Correlation between the two ranking criteria (LC₅₀ and MLCR) was high ($r_s = 0.768$, $P < 0.0001$, $n = 22$), though somewhat lower than the correlations observed within lepidopteran species ($r_s = 0.784$ – 0.944). In the lists of isolates ranked by each variable as 10th or higher across Lepidoptera species, the top three isolates (1200, 1083, and 1084) and their ranks were identical across the two ranking systems, and 16 of 20 isolates (80%) ranked 10th or higher by one system were also ranked in the top 10 by the alternative systems (Table 10). When closely related isolates were assigned the rank of the most highly ranked member of the group (as in Table 8), 16 of 18 isolates/isolate groups (89%) ranked 9th or higher were included in both rankings (Table 10).

3.3. Data distributions

Results of the Shapiro–Wilk test and examinations of the normal quantile plots revealed little evidence of non-normal distributions of the three probit regression variables from the replicated screening bioassays of strain GHA. (Table 11). The intercepts and log LC₅₀ values from the corn earworm assays were notable exceptions ($P \leq 0.065$), exhibiting distributions that were markedly asymmetrical (skewed).

3.4. Comparative virulence assays

Results of the comparative virulence assays are presented in Table 12. Isolates are ranked in the table by LC₅₀. These assays revealed statistically significant differences among the most highly ranked isolates tested against five lepidopteran pests. The parametric ANOVA followed by the Tukey test for all pairwise means comparisons and the nonparametric Kruskal–Wallis test followed by sequential Bonferroni comparisons produced nearly identical results. Isolate 1200 was the most virulent fungus against diamondback moth, fall armyworm, and beet armyworm. This isolate was significantly more virulent than commercial strain GHA against all pests except corn earworm and was as virulent as GHA against this pest.

Few statistically significant differences in probit regression coefficients (slopes) were detected among fungal isolates within pest species (Table 12). Slopes across all assays ranged from approximately 1.2 to 2.6. Within pest species, there was generally a negative correlation between log LC₅₀ and slope (positive correlation between virulence and slope); however, with the exception of fall armyworm, the correlations were not statistically significant (Table 12).

3.5. Tests against other Lepidoptera species

Based on a limited number of assays, isolate 1200 exhibited high virulence against imported cabbageworm (LC₅₀ = 12 conidia/mm²) and approximately 10–20-fold lower virulence against cabbage looper (LC₅₀ = 125 conidia/mm²) and black cutworm (LC₅₀ = 273 conidia/mm²) (Table 13). ANOVA indicated isolate 1200 was significantly more virulent than strain GHA against each of the three pests ($F_{[1, 2]} = 73.9$, $P = 0.013$; $F_{[1, 3]} = 37.1$, $P = 0.009$; $F_{[1, 2]} = 24.2$, $P = 0.039$, respectively).

Table 9

Mean rankings of the 23 most virulent isolates of *Beauveria bassiana* based on median lethal rates (LC₅₀) and median lethal concentration ratios (MLCR) across five lepidopteran pest species.

Isolate ^a	Diamondback moth	European corn borer	Corn earworm	Fall armyworm	Beet armyworm	Mean LC ₅₀ rank ^b	Mean MLCR rank ^b
<i>Ranks based on LC₅₀ and (MLCR)^c</i>							
1200	1 (2)	18 (16)	3 (1)	1 (1)	1 (1)	4.8	4.2
1083	10 (6)	30 (19)	6 (6)	4 (2)	5 (6)	11.0	7.8
1084	3 (1)	31 (21)	15 (8)	7 (3)	3 (7)	11.8	8.0
1153	7 (8)	26 (24)	5 (13)	6 (12)	23 (23)	13.4	16.0
1082	5 (3)	22 (22)	26 (19)	11 (6)	8 (3)	14.4	10.6
SA-8	7 (11)	24 (30)	12 (11)	16 (24)	16 (13)	15.0	17.8
1512	26 (36)	4 (1)	20 (31)	8 (16)	19 (16)	15.4	20.0
1075	6 (10)	27 (34)	4 (4)	17 (30)	24 (14)	15.6	18.4
SC15	24 (22)	40 (39)	2 (3)	2 (4)	14 (4)	16.4	14.4
SC19	23 (20)	37 (36)	13 (7)	3 (7)	6 (12)	16.4	16.4
2638	15 (14)	29 (25)	8 (5)	18 (8)	13 (17)	16.6	13.8
1009	27 (24)	6 (3)	38 (26)	12 (14)	9 (2)	18.4	13.8
1513	30 (36)	5 (8)	41 (40)	5 (8)	12 (11)	18.6	20.6
1966	2 (4)	20 (29)	28 (25)	27 (28)	17 (21)	18.8	21.4
3288	4 (5)	35 (34)	1 (2)	38 (37)	21 (20)	19.8	19.6
1022	38 (36)	11 (6)	29 (14)	10 (22)	11 (15)	19.8	18.6
SE6	19 (17)	33 (28)	19 (29)	23 (17)	6 (8)	20.0	19.8
1055	18 (16)	21 (26)	14 (12)	30 (13)	22 (25)	21.0	18.4
1023	28 (28)	2 (2)	40 (34)	34 (31)	3 (5)	21.4	20.0
GMB6B	42 (41)	8 (10)	31 (37)	25 (11)	2 (10)	21.6	21.8
GHA	13 (–)	32 (–)	10 (–)	31 (–)	25 (–)	22.2	–
2559	17 (18)	28 (27)	36 (18)	15 (15)	18 (19)	22.8	19.4
1060	37 (32)	1 (4)	43 (37)	22 (27)	14 (9)	23.4	21.8

^a Isolates ordered by mean LC₅₀ rank.

^b Spearman's coefficient of rank correlation (r_s) between the mean LC₅₀ and MLCR ranks = 0.768.

^c Ranks from data presented in Tables 3–7.

Table 10

Fungal isolates or isolate groups ranked tenth or higher across five lepidopteran pest species based on median lethal concentrations (LC₅₀) and median LC ratios (MLCR)^a.

Isolates Ranking based on		Isolates or isolate groups ^b Ranking based on		
LC ₅₀ ^{c,d}	MLCR ^{c,d}	LC ₅₀ ^{c,d}	MLCR ^{c,d}	Mean of LC ₅₀ and MLCR ranks ^d
1 – 1200 (4.8)	1 – 1200 (4.2)	1 – 1200 (4.8)	1 – 1200 (4.2)	1 – 1200 (4.5)
2 – 1083 (11.0)	2 – 1083 (7.8)	2 – 1083/84/82 (11.0)	2 – 1083/84/82 (7.8)	2 – 1083/84/82 (9.4)
3 – 1084 (11.8)	3 – 1084 (8.0)	3 – 1153 (13.4)	3 – 2638 (13.8)	3 – 1153 (14.7)
4 – 1153 (13.4)	4 – 1082 (10.6)	4 – SA8 (15.0)	4 – 1009/22 (13.8)	4 – 2638 (15.2)
5 – 1082 (14.4)	5 – 2638 (13.8)	5 – 1512/13 (15.4)	5 – SC15/19 (14.4)	5 – SC15/19 (15.4)
6 – SA8 (15.0)	5 – 1009 (13.8)	6 – 1075 (15.6)	6 – 1153 (16.0)	6 – 1009/22 (16.1)
7 – 1512 (15.4)	7 – SC15 (14.4)	7 – SC15/19 (16.4)	7 – SA8 (17.8)	7 – SA8 (16.4)
8 – 1075 (15.6)	8 – 1153 (16.0)	8 – 2638 (16.6)	8 – 1075 (18.4)	8 – 1075 (17.0)
9 – SC15 (16.4)	9 – SC19 (16.4)	9 – 1009 (18.4)	8 – 1055 (18.4)	9 – 1512/13 (17.7)
9 – SC19 (16.4)	10 – SA8 (17.8)	10 – 1966 (18.8)	10 – 2559 (19.4)	10 – 3288 (19.7)
21 – GHA (–)		17 – GHA (–)		10 – 1055 (19.7)

^a Ranks based on data presented in Table 9.

^b Groups of closely related isolates are assigned the rank of the highest ranking isolate in the group; the highest ranked isolate is listed first.

^c Isolates or isolate groups in bold type were ranked in the top 10 by only one of the alternative ranking systems (LC₅₀ versus MLCR).

^d Numbers in parentheses are the means of the ranks across lepidopteran species for each isolate or for the highest ranked isolate of a group.

Table 11

Results of the Shapiro–Wilk test for normality of slopes, intercepts, and median lethal concentration values^a from probit analyses of replicated assays of *Beauveria bassiana* strain GHA against five species of lepidopteran larvae.

Pest species	No. of assays	Slopes	Intercepts	CV ratio ^b	Log LC ₅₀
Diamondback moth	28	$W = 0.945, P = 0.13$	$W = 0.935, P = 0.068$	1.2 (41/35)	$W = 0.970, P = 0.55$
European corn borer	33	$W = 0.967, P = 0.40$	$W = 0.967, P = 0.39$	3.4 (124/37)	$W = 0.971, P = 0.51$
Corn earworm	35	$W = 0.959, P = 0.21$	$W = 0.819, P < 0.0001$	0.6 (25/42)	$W = 0.942, P = 0.065$
Fall armyworm	33	$W = 0.967, P = 0.67$	$W = 0.966, P = 0.37$	6.3 (170/27)	$W = 0.971, P = 0.51$
Beet armyworm	15	$W = 0.951, P = 0.54$	$W = 0.964, P = 0.76$	2.1 (69/33)	$W = 0.957, P = 0.63$

^a Results of tests of the hypothesis that the data are normally distributed.

^b Coefficient of variation, CV, of intercepts/CV of slopes.

4. Discussion

B. bassiana was pathogenic against larvae of all Lepidoptera species tested, and numerous highly virulent isolates were identified.

The exceedingly low LC₅₀s (<5 conidia/mm²) associated with a number of isolates are likely a technical artifact of the assay system, which provided conditions highly favorable to inoculation and infection. Newly molted second-instar larvae were exposed

Table 12Direct comparisons of high-ranking isolates of *Beauveria bassiana* against five species of Lepidoptera.

Isolate (% viability) ^a	Number of assays (tests) ^b	LC ₅₀ (log LC ₅₀ ± SE) ^c	Slope ± SE ^c	% Control mortality ± SE	Kruskal–Wallis chi-square/ANOVA <i>F</i> test results Correlation coefficient (<i>r</i>) ^d
<i>Diamondback moth</i>					
1200 (91)	6 (2)	1.1 (0.05 ± 0.10) a A	2.25 ± 0.22 ab	8.3 ± 2.1	LC ₅₀ $\chi^2_{[5]} = 28.6$, $P < 0.0001$ Log LC ₅₀ $F_{[5, 29]} = 25.7$, $P < 0.0001$ Slope $F_{[5, 29]} = 3.4$, $P = 0.015$ $r = -0.470$, $P = 0.35$
1082 (92)	6 (2)	2.8 (0.45 ± 0.06) b B	2.09 ± 0.32 ab		
1084 (92)	6 (2)	4.9 (0.69 ± 0.11) b BC	2.58 ± 0.32 a		
1966 (84)	6 (2)	6.8 (0.84 ± 0.10) bc CD	1.15 ± 0.14 b		
GHA (93)	6 (2)	14.2 (1.15 ± 0.04) c D	1.90 ± 0.33 ab		
3288 (90)	6 (2)	14.8 (1.17 ± 0.09) c D	1.67 ± 0.20 ab		
<i>European corn borer</i>					
1009 (92)	6 (2)	7.6 (0.88 ± 0.08) a A	1.38 ± 0.15 ab	9.0 ± 1.2	LC ₅₀ $\chi^2_{[5]} = 26.0$, $P < 0.0001$ Log LC ₅₀ $F_{[5, 29]} = 53.7$, $P < 0.0001$ Slope $F_{[5, 29]} = 3.3$, $P = 0.018$ $r = -0.475$, $P = 0.34$
1512 (92)	6 (2)	9.9 (1.00 ± 0.16) a A	1.27 ± 0.14 b		
1060 (93)	6 (2)	10.9 (1.04 ± 0.07) a A	2.40 ± 0.32 a		
1023 (91)	6 (2)	14.7 (1.17 ± 0.07) a A	1.71 ± 0.14 ab		
1200 (92)	6 (2)	139.2 (2.14 ± 0.15) b B	1.66 ± 0.41 ab		
GHA (90)	6 (2)	1913 (3.28 ± 0.19) c C	1.03 ± 0.25 b		
<i>Corn earworm</i>					
SC15 (89)	4 (2)	1.0 (0.001 ± 0.21) a A	1.92 ± 0.62 a	3.2 ± 0.1	LC ₅₀ $\chi^2_{[5]} = 3.4$, $P = 0.64$ Log LC ₅₀ $F_{[5, 17]} = 1.6$, $P = 0.22$ Slope $F_{[5, 17]} = 0.65$, $P = 0.67$ $r = -0.682$, $P = 0.14$
1200 (88)	4 (2)	1.4 (0.16 ± 0.07) a A	1.73 ± 0.19 a		
1075 (94)	4 (2)	1.8 (0.25 ± 0.21) a A	1.23 ± 0.27 a		
GHA (86)	4 (2)	2.2 (0.34 ± 0.15) a A	1.97 ± 0.66 a		
1083 (94)	4 (2)	3.2 (0.50 ± 0.15) a A	1.19 ± 0.36 a		
3288 (86)	4 (2)	4.3 (0.63 ± 0.23) a A	1.19 ± 0.40 a		
<i>Fall armyworm</i>					
1200 (93)	9 (3)	15.1 (1.18 ± 0.08) a A	2.60 ± 0.12 a	2.8 ± 1.1	LC ₅₀ $\chi^2_{[5]} = 37.1$, $P < 0.0001$ Log LC ₅₀ $F_{[5, 46]} = 57.9$, $P < 0.0001$ Slope $F_{[5, 46]} = 4.1$, $P = 0.004$ $r = -0.877$, $P = 0.022$
1083 (96)	9 (3)	169.0 (2.23 ± 0.08) b B	1.62 ± 0.18 ab		
1513 (97)	9 (3)	360.1 (2.56 ± 0.13) bc BC	1.17 ± 0.16 b		
SC15 (94)	9 (3)	432.6 (2.64 ± 0.07) c C	1.64 ± 0.18 ab		
1153 (89)	9 (3)	573.1 (2.76 ± 0.09) c C	1.50 ± 0.26 b		
GHA (90)	9 (3)	1802 (3.26 ± 0.07) d D	1.37 ± 0.27 b		
<i>Beet armyworm</i>					
1200 (91)	5 (2)	2.5 (0.40 ± 0.07) a A	2.35 ± 0.52 a	4.9 ± 3.4	LC ₅₀ $\chi^2_{[4]} = 21.1$; $P = 0.0003$ Log LC ₅₀ $F_{[4, 19]} = 38.6$; $P < 0.0001$ Slope $F_{[4, 19]} = 1.5$; $P = 0.254$ $r = -0.396$, $P = 0.51$
1084 (92)	5 (2)	5.1 (0.71 ± 0.11) b B	1.40 ± 0.27 a		
GMB6B (90)	5 (2)	17.7 (1.25 ± 0.07) c C	1.54 ± 0.12 a		
1023 (89)	5 (2)	20.0 (1.30 ± 0.07) c C	1.51 ± 0.25 a		
GHA (90)	5 (2)	115.1 (2.06 ± 0.15) d D	1.72 ± 0.23 a		

^a Mean percent viability recorded across replicate assays.^b Number of replicate assays conducted in two or three tests, each test conducted on a different occasion (randomized complete block design, blocked over time).^c Means (± standard errors) within columns within pest species not followed by same letter are significantly different (sequential Bonferroni test following Kruskal–Wallis test [lower case letters] versus Tukey HSD test following parametric ANOVA [upper case letters]; Tukey test comparing slopes; experimentwise alpha = 0.05).^d Test results from nonparametric chi-square versus parametric *F* tests; correlation coefficient (*r*) between log LC₅₀ and probit regression slope.

for 24 h to a spongy, hydrophilic substrate (moist filter paper) contaminated with hydrophobic conidia, and secondary pickup of conidia was likely an important factor. Nevertheless, the generally high susceptibility to *B. bassiana* displayed by most species was not anticipated. Tests of the different pest species were conducted on different occasions, precluding rigorous statistical comparisons, but among the most extensively tested insects, beet armyworm and corn earworm appeared most susceptible, with most isolates (90% and 74%, respectively) exhibiting LC₅₀s of <100 conidia/mm². European corn borer and diamondback moth displayed intermediate susceptibility, with 42% and 30% of isolates having LC₅₀s less than 100 conidia/mm². Fall armyworm was clearly the least susceptible of the most extensively tested insects; an LC₅₀ of less than 100 conidia/mm² was recorded for only a single isolate (1200).

The reason for the generally low susceptibility of fall armyworm is unknown. The duration of the second larval instar of this species is short (1.7 days at 25 °C) (Pitre and Hogg, 1983), and the molt to third instar may have interfered with fungal penetration (Vey and Fargues, 1977; Vandenberg et al., 1998). On the other hand, the duration of the second larval instar of beet armyworm is similar (1.5 days at 30 °C) (Fye and McAda, 1972), yet this insect was highly susceptible.

In a previous study in which a comparable number of fungal isolates were screened against whiteflies (Wraight et al., 1998), it was observed that lethal concentration ratios (referred to as potency ratios in that paper) were as variable as LC₅₀s, and thus of-

fered little or no advantage as a tool for detecting differences in virulence among isolates. It was further argued that this was not unexpected since such ratios were developed for assays of chemical toxins exhibiting a common mode of action and that changes in environmental or host conditions from assay to assay could not be expected to have the same effect on all strains of a pathogenic microorganism. The same conclusion can be drawn from the data of the present study. If one examines the average coefficients of variation (CVs) of the LC₅₀s (linear scaled) versus MLCRs of isolates in groups of related strains, there are clearly no trends (Table 14). It follows that if LC₅₀ values from assay to assay are no more variable than the LC₅₀ ratios (even over prolonged periods of time), there is little to recommend use of the ratios in the screening of diverse isolates of fungal pathogens. In view of the considerable effort required to include a “standard” in every test, it might be argued that greater benefit would derive from applying those resources toward increased replication in assaying the unknowns or toward testing a greater number of unknowns. This conclusion is supported by the high correlations observed between the isolate rankings based on LC₅₀ versus MLCR in the series of assays conducted both within and across pest species. The two ranking systems tended to identify the same isolates as highly virulent.

Mycotech Corporation assessed sporulation capacity of the selected isolates, and isolates that sporulated poorly on artificial substrates were not included in the screening program. Important exceptions, however, were the closely related isolates 1082,

Table 13Virulence of selected *Beauveria bassiana* isolates against black cutworm, imported cabbageworm, and cabbage looper.

Isolate (% viability) ^a	Assays (n)	LC ₅₀ (log LC ₅₀ ± SE) rank ^b	Slope ± SE	Median LC ratio ± SE (rank) ^b	% Mortality (dose) C ± SE ^c
<i>Black cutworm</i>					
1084 (98)	1	223 (2.35 ± –) 1	1.39 ± –	19.6 ± – (1)	92 (2341) 0 ± –
1200 (95)	3	273 (2.44 ± 0.03) 2	2.35 ± 0.42	14.2 ± 2.4 (2)	94 (1387) 1 ± 0.7
1083 (96)	1	350 (2.54 ± –) 3	1.89 ± –	12.5 ± – (3)	94 (2396) 0 ± –
1153 (95)	1	1070 (3.03 ± –) 4	2.21 ± –	5.3 ± – (4)	72 (1924) 0 ± –
1055 (92)	1	1094 (3.04 ± –) 5	1.70 ± –	4.0 ± – (6)	83 (2961) 0 ± –
1966 (91)	1	1350 (3.13 ± –) 6	1.48 ± –	4.2 ± – (5)	76 (3419) 0 ± –
2559 (95)	2	2127 (3.33 ± 0.27) 7	1.74 ± 0.96	2.7 ± 2.2 (7)	88 (2854) 0 ± 0
1009 (93)	1	2347 (3.37 ± –) 8	1.64 ± –	1.9 ± – (8)	53 (2699) 0 ± –
GHA (94)	5	3504 (3.54 ± 0.08) 9	1.55 ± 0.28	–	75 (4778) 1 ± 0.6
3288 (88)	1	5886 (3.77 ± –) 10	0.89 ± –	0.4 ± – (9)	25 (1165) 0 ± –
<i>Imported cabbageworm</i>					
1200 (96)	3	13 (1.11 ± 0.06)	1.92 ± 0.37	2.3 ± 0.2	93 (209) 17 ± 9.6
GHA (93)	2	29 (1.46 ± 0.04)	1.30 ± 0.05	–	96 (699) 17 ± 9.6
<i>Cabbage looper</i>					
1200 (94)	3	131 (2.12 ± 0.08)	1.75 ± 0.33	5.5 ± 1.7	96 (1512) 1 ± 0.7
GHA (95)	3	653 (2.82 ± 0.17)	1.73 ± 0.23	–	94 (5171) 1 ± 0.7

^a Mean percent viability recorded across the replicate assays.^b Spearman's coefficient of rank correlation (r_s) between LC₅₀ and median LC ratio for black cutworm = 0.983.^c Maximum% mortality (excluding 100% values); dose (viable conidia/mm²) producing indicated mortality; C = mean% control mortality.

1083, and 1084. These isolates were inefficient conidia producers under commercial laboratory conditions, producing only 10–20% as many conidia as GHA (Jaronski, unpublished data), but were tested because preliminary assays had identified their exceptional virulence and host range. This low sporulation capacity may limit the commercial potential of these otherwise very promising isolates. One final note on this group of isolates: it did not surprise us to find that isolates 1083 and 1084 were closely related, as they were both collected on the same date from gypsy moth in Allegany County, Maryland, USA. Isolate 1082, however, is reported to have originated from a sawfly larva in British Columbia, Canada. This suggests a possible contamination problem. Genetic fingerprinting of all the most promising isolates included in this study is warranted.

The extraordinarily high virulence and broad host range of isolate 1200 was an unexpected discovery. Commercial strain GHA is widely recognized as a generalist pathogen, with demonstrated virulence against a broad range of insect pests from different insect orders, yet isolate 1200 proved to have virulence greater than or equal to GHA against all eight lepidopteran species tested. Based on a limited number of tests, black cutworm was the least susceptible to this isolate of all the lepidopteran species we tested, with an LC₅₀ of 273 conidia/mm² (Table 13). In comparison, the LC₅₀ of European corn borer, the least susceptible of the principal insects tested, was only approximately 100 conidia/mm² (Tables 4 and 9). Nevertheless, isolate 1200 was substantially more virulent against black cutworm than commercial strain GHA, and appeared also more virulent than several of the other isolates ranked highly against other species (e.g., isolates 3288 and 1009). Only isolates 1083 and 1084, which ranked second to isolate 1200 in the host range evaluations, exhibited pathogenicity comparable to 1200 against this pest. Isolate 1200 was also more virulent than GHA against cabbage looper and imported cabbageworm.

The finding of a direct, though in most cases insignificant correlation between probit regression slope and virulence was a positive result in that it enabled us to focus on LC₅₀ in comparing virulence of the various isolates. In the direct comparisons of top-ranked isolates, we observed no instance in which an isolate exhibiting significantly greater virulence than another isolate (based on LC₅₀) was associated with a significantly lower slope (Table 10). Though observed in this study, this relationship should not be assumed universal for all host–pathogen systems; slopes of probit regression lines, which greatly impact extreme LC values such as LC₉₅S,

must always be taken into account when assessing the microbial biological control potential of entomopathogenic fungi.

Slopes recorded for some isolates were unexpectedly high. Pathogenic microorganisms tend to act independently rather than synergistically in establishing host infection, and this mode of

Table 14Comparison of Coefficients of variation (CVs) of median lethal concentrations (LC₅₀) and median lethal concentration ratios (MLCR) of five isolates/isolate groups of *Beauveria bassiana* against five species of Lepidoptera.

Pest species	CV of LC ₅₀ s (%)	CV of MLCRs (%)	Assays (n)
<i>Corn isolates (isolates 1003–1023)</i>			
Diamondback moth	65	100	23
European corn borer	110	135	36
Corn earworm	153	137	26
Fall armyworm	99	56	25
Beet armyworm	50	102	7
Mean	95.4	106.0	
<i>Isolate 1200</i>			
Diamondback moth	115	27	3
European corn borer	97	99	4
Corn earworm	86	95	3
Fall armyworm	54	131	3
Beet armyworm	54	49	3
Mean	62.1	80.2	
<i>Isolates SC15/19</i>			
Diamondback moth	34	99	4
European corn borer	85	56	5
Corn earworm	44	95	5
Fall armyworm	85	85	7
Beet armyworm	107	119	5
Mean	71.0	90.8	
<i>Isolates 1512/13</i>			
Diamondback moth	27	0	3
European corn borer	79	142	6
Corn earworm	180	71	4
Fall armyworm	132	43	6
Beet armyworm	24	33	4
Mean	88.4	57.8	
<i>Isolates 1082/83/84</i>			
Diamondback moth	69	137	7
European corn borer	44	44	6
Corn earworm	103	67	7
Fall armyworm	57	78	9
Beet armyworm	65	107	7
Mean	67.6	86.6	
Grand mean	76.9	84.3	

action translates to a theoretical maximum probit regression slope of approximately 2.0 (Meynell, 1957; Huber and Hughes, 1984). It may be that the average slopes substantially >2 such as associated with isolate 1200 and other highly virulent isolates (Table 12) were an artifact of our assay protocol. Highly virulent isolates are capable of producing infections at exceedingly low doses, but because survival time is inversely related to dose, such infections may require a prolonged period to effect host mortality. Termination of our assays after 6 days may have created a bias toward higher slopes.

Excellent fit of the log LC_{50} data to the normal distribution in four out of five cases (Table 11) was another unexpected observation. LC_{50} estimates are derived as the ratio of two variables that are usually normally distributed: the intercept and slope of linear dose–response regression lines (Hinkley, 1969). Ratios of normally distributed variables are generally described as fitting a non-normal, heavy-tailed distribution known as the Cauchy–Lorentz distribution (also called the normal ratio distribution). However, this outcome is largely theoretical; distributions of ratios derived from variables with approximately normal distributions, truncated normal distributions, or normal distributions with unequal variances and with varying degrees of correlation—common characteristics of real-world biological data—are more variable and complex than Cauchy–Lorentz and may be approximately normal under certain conditions (Hinkley, 1969; Hayya et al., 1975; Atchley et al., 1976). The potential for significant non-normality of ratio data must be taken into consideration in any application of parametric ANOVA. In a simulation study, Anderson and Lydic (1977) observed that the power of an ANOVA of ratios was highly dependent upon the ratio of the CVs of the numerator and denominator variables and concluded that ANOVA of ratios was acceptable under the condition that the CV of the denominator was $\leq$ the CV of the numerator. The results from our study support this observation. Strain GHA LC_{50} values identified by the Shapiro–Wilk test as fitting normal distributions were associated with intercept/slope CV ratios >1 (Table 11). Considering the inherently low slopes of probit regression lines associated with infectious agents, one would expect that lower slope versus intercept CVs would be the norm. Robertson et al. (1995) present LC_{50} s from extensively replicated bioassays of *Bacillus thuringiensis* against Colorado potato beetle (83 assays) and diamondback moth (37 assays). When log transformed and subjected to Shapiro–Wilk analysis, these data also show good fit to the normal distribution ($W = 0.977$, $P = 0.45$ and $W = 0.969$, $P = 0.47$, respectively).

Heavy-tailed (leptokurtic) distributions tend to be outlier prone (Zimmerman, 2001; Brody et al., 2002). By increasing variance, outliers can markedly reduce the power of parametric ANOVA to detect significant differences. For such data, nonparametric ranking tests can provide substantially greater power and are highly recommended (Zimmerman, 2001). In anticipation of this form of non-normality of log LC_{50} data, we initially planned to conduct all comparisons of the LC_{50} s associated with different fungal strains using the nonparametric tests described in Section 2.5. Upon observing good fit of the log LC_{50} data to the normal distribution, however, we became interested in comparing results of the nonparametric analyses to those from standard parametric ANOVA/Tukey tests. The ultimate finding of similar results from the two analyses (with no apparent gain in power from application of the nonparametric approach) argues strongly for acceptance of parametric tests for comparisons of LC_{50} s unless there is evidence of a substantial departure from normality.

In light of these results, we suggest that a reasonable approach for analysis of virulence data in the form of replicated log LC_{50} s or ST_{50} s (median survival times) is to examine normal quantile (Q–Q) plots, histograms, etc. and conduct one or more tests for normality of the data, recognizing that such tests have only limited power to reject the hypothesis of normality when samples are small (<9

according to D'Agostino et al., 1990). Indeed, production of replicated LC_{50} s for more rigorous testing of the distribution of data generated by a specific bioassay protocol is one argument in favor of including a standard strain in all tests. In cases of small sample sizes among multiple treatments, normality for the overall analysis can be tested by generating standardized deviates within each sample, pooling the deviates across samples, and then testing them for normality via the Shapiro–Wilk or other tests (Sokal and Rohlf, 1995). If these methods reveal no serious deviations from normality, application of a standard ANOVA and a stringent multiple comparison test that controls experimentwise alpha should be considered reliable. Parametric ANOVA is generally robust to departures from normality (Sokal and Rohlf, 1995; Zar, 1999) and, as related above, is a conservative test for data from heavy-tailed distributions. If non-normality is detected or sample sizes are too small to enable confident characterization of the distribution, a nonparametric ranking test should be employed, at least as a confirmation of ANOVA results (and as related above, if outliers are present, the nonparametric test may prove more powerful than the ANOVA). With respect to multiple comparisons following nonparametric tests, the sequential Bonferroni procedure exhibits reasonably good power and is extremely easy to conduct. Results from this method were nearly identical to those from the parametric Tukey HSD test in the present study. In a Monte Carlo study comparing five treatments, Seaman et al. (1991) also observed similar results from application of the parametric Tukey HSD versus sequential Bonferroni tests.

Significant outlier values were notably absent in the data from our five-dose comparative virulence assays. This result supports the suggestion by Robertson and Preisler (1992) that reasonably good estimates of LC_{50} may be obtainable from assays employing as few as 60 subjects (the minimum number of larvae used in our replicate assays). The strongly skewed distribution of probit regression intercepts and LC_{50} values from the corn earworm assays (Table 12) appears to be a reflection of the extremely high susceptibility of this insect to strain GHA (nearly 70% of the LC_{50} estimates were ≤ 10 conidia/mm²). Truncated distributions are common in biology. In this case, LC_{50} cannot be <0 . Though lacking outliers, both the LC_{50} s and MLCRs associated with fungi tested repeatedly against an insect host proved highly variable (Table 14). In addition to the high coefficients of variation (CVs) shown in Table 14, the CVs of strain GHA LC_{50} s were 43%, 67%, 97%, 120%, and 163% against beet armyworm, diamondback moth, fall armyworm, European corn borer and corn earworm, respectively. Robertson et al. (1995) also noted high variability in LC_{50} estimates from replicated *B. thuringiensis* assays (see above) and concluded that this was a reflection of high natural variation in susceptibility of insect populations. In addition to this “high natural variation,” experimental errors associated with the preparation and quantification of pathogen suspensions and delivery of precise doses must also be recognized as important factors.

Some discussion is warranted to explain our selection of the above-described ANOVA-based analysis over the more traditional approach involving combination of replicate assays into single probit regressions for estimating mean LC_{50} s with fiducial limits. The decision relates primarily to limitations in experimental design and availability of methods for post hoc comparisons. The assays in our screening tests (Tables 3–7) were obviously not conducted in a randomized complete block design. We usually could not conduct more than 6–8 assays on a single day. Thus, while the data from our replicate assays could have been combined and fiducial limits generated, these limits would be meaningless with respect to any pairwise comparisons among isolates. Analysis of screening data is commonly constrained by this problem. Ultimately, researchers can do little more than identify what appear to be the best candidates and then compare these candidates in head-to-head tests.

The comparative virulence assays (Table 12) were conducted in randomized complete block designs (within each pest species), and analysis of combined assays using a statistical program designed to handle block as a factor (e.g., a generalized linear model with a probit link function) could provide unbiased estimates of LC_{50} s and slopes and meaningful fiducial limits. However, comparing any confidence intervals generated without consideration of the number of intended comparisons is not the equivalent of a rigorous statistical test; this procedure does not control the experimentwise error rate. Beyond this, there are general tests for comparing probit lines (testing for equality of slopes and intercepts), but when slopes are unequal (significant isolate $\times$ dose interaction), these comparisons cannot test for differences in LC values, and there are few options for multiple comparisons (Robertson et al. (2007)). We recommend the ANOVA approach based on its combined attributes of rigor and simplicity. We have provided $\log LC_{50}$ values with standard errors and numbers of replicates (assays) from which conventional 95% confidence intervals based on normal sampling theory are readily calculable if desired.

This study exemplifies the potential to find strains of common entomopathogenic fungi with combined traits of high virulence and broad host range and indicates potential for development of microbial control agents not only for management of key pests but also of important pest complexes. We were surprised to have identified such an interesting isolate as *B. bassiana* 1200 through a screening of only a relatively few isolates. Indeed, far more extensive efforts might be less fortunate. Nevertheless, this work confirms that screening of fungal isolates remains a valid undertaking in the field of applied insect mycopathology.

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